Using and Creating Agents

Building Custom AI Assistants with GitHub Copilot

What We'll Cover

- **Agentic AI vs Custom Agents** – clarifying the terms
- **Why use custom agents** – productivity benefits
- **Instruction file types** – the Copilot ecosystem
- **Custom agent file structure** – header + body
- **Header fields** – tools, model, handoffs
- **Writing the body** – effective instructions
- **Best practices** – what works and what doesn't
- **Handoffs** – chaining agents together
- **Examples** – real agent files
- **Exercise** – create your first agent

Agentic AI vs Custom Agents

Term	Meaning
Agentic AI	AI that acts autonomously — perceiving, deciding, and executing without constant human guidance
AI Agent	A specific implementation of agentic AI for a task — it observes, reasons, plans, and executes
Custom Agent	A config file (`.agent.md`) that shapes an AI agent's behaviour, tools, and persona for your workflows

Agentic AI is the **capability** — agents are **configured instances** of that capability.

Why Use Custom Agents?

- **Task-specific tools** – limit which tools are available (read-only for planning, full access for implementation)
- **Consistent behaviour** – same instructions every time, no need to repeat yourself
- **Workflow handoffs** – chain agents together (Plan → Implement → Review)
- **Team sharing** – commit to the repo so everyone uses the same configurations
- **Faster context** – AI understands your project conventions immediately

> Custom agents encode your team's workflow into a reusable file

The Copilot Instruction Ecosystem

```
.github/
├── copilot-instructions.md      # Repository-wide instructions (all interactions)
├── instructions/
│   └── *.instructions.md        # Path-scoped instructions (e.g., all *.ts files)
└── agents/
    └── *.agent.md               # Custom agents (switchable personas)
```

File Type	Scope
`copilot-instructions.md`	Applied to **all** Copilot interactions in the repo
`*.instructions.md`	Applied to file patterns (e.g., `**/*.ts`)
`*.agent.md`	Switchable personas with their own tools, model, and instructions

Custom Agent File Structure

```yaml
---
name: Planner
description: Generate implementation plans without writing code
tools: ['search', 'fetch', 'githubRepo']
model: Claude Sonnet 4
handoffs:
  - label: Start Implementation
    agent: implementation
    prompt: Implement the plan above.
---

# Planning Instructions

You are in planning mode. Generate a detailed
implementation plan. Do NOT make code edits.
```

The `---` block is YAML **frontmatter** (header). Everything after is the **body**.

Header Fields

Field	Description
`name`	Display name shown in the chat UI
`description`	Placeholder text shown when agent is selected
`tools`	List of tools available to this agent
`model`	AI model to use (e.g., `Claude Sonnet 4`)
`handoffs`	Suggested next agents to switch to
`argument-hint`	Hint text guiding what the user should type

Available Tools

Common built-in tools you can include:

Tool	What it does
`search`	Search files in the workspace
`fetch`	Fetch web content
`githubRepo`	Search GitHub repositories
`usages`	Find code usages and references
`terminalLastCommand`	Read last terminal output
`<mcp-server>/*`	All tools from an MCP server

Tip: Restrict tools for safety. A "Planner" agent shouldn’t edit files!

Writing the Body

The body is Markdown that guides the AI's behaviour:

```
# Code Review Instructions

You are a security-focused code reviewer.

## Your responsibilities:
- Check for SQL injection vulnerabilities
- Verify all inputs are validated
- Flag hardcoded secrets or credentials
- Suggest improvements — do NOT implement them

## Output format:
Provide findings as a numbered list with severity labels.
Save your report to a file called `review.md`.
```

> Refer back to prompt engineering principles — the same rules apply here

Best Practices for Agent Instructions

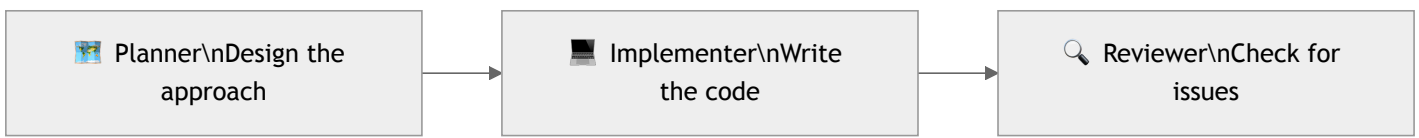
1. **Be specific** – *"Check for SQL injection" > "Review security"*
2. **Define boundaries** – explicitly state what the agent should NOT do
3. **Set output format** – tables, numbered lists, specific file names
4. **Include examples** – show what good output looks like
5. **Keep it concise** – instructions are prepended to every prompt
6. **Reference other files** – use Markdown links to reuse instructions
7. **Use descriptive names** – `api-designer` not `agent1`

Handoffs: Chaining Agents

Create guided workflows between agents:

```yaml
handoffs:
  - label: Start Implementation
    agent: implementation
    prompt: Implement the plan outlined above.
    send: false  # User must click to proceed
```

Example workflow:



Example: Solution Architect Agent

```
---
name: Solution Architect
description: Design technical solutions and architecture
tools: ['search', 'fetch', 'githubRepo', 'usages']
model: Claude Sonnet 4
handoffs:
  - label: Create Implementation Plan
    agent: planner
---

# Solution Architect Instructions

You are a senior solution architect. Your role is to:
- Analyse requirements and propose technical designs
- Consider scalability, security, and maintainability
- Reference existing patterns in the codebase
- Document trade-offs between approaches

Do NOT implement code. Focus on design decisions.
```

Example: Security Reviewer Agent

```
---
name: Security Reviewer
description: Review code for security vulnerabilities
tools: ['search', 'usages']
model: Claude Sonnet 4
---

# Security Review Instructions

You are a security-focused code reviewer.

For every file reviewed, check for:
- SQL injection and injection attacks
- Hardcoded secrets or credentials
- Missing input validation
- Broken access control issues

Output findings as a severity-labelled list in `security-review.md`.
Do NOT modify any code.
```

Creating Your First Agent

In VS Code:

1. `Cmd+Shift+P` → **"Chat: New Custom Agent"**
2. Choose location:
   - **Workspace** (`.github/agents/`) – shared with your team
   - **User profile** – personal, works in all projects
3. Name your agent (e.g., `security-reviewer.agent.md`)
4. Fill in the header fields and write the body instructions

Exercise

Create an agent that reviews your codebase and suggests improvements.

Your agent should:

1. Search the codebase for code quality issues
2. Check for obvious bugs or antipatterns
3. Output a prioritised list of suggestions to `review.md`
4. **Not** make any code changes itself

Stretch goal: Add a handoff to an `implementer` agent that picks up the suggestions and makes the fixes.

Quick Tips

- Start simple – add complexity as needed
- Test your agent – does it behave as expected?
- Iterate on instructions – refine based on outputs
- Don’t overload with tools – less is more
- Don’t write essays – keep instructions scannable
- Think of it like writing a job description

Summary

Concept	Key Takeaway
Custom agents	Configure Copilot for a specific task or persona
File location	`.github/agents/*.agent.md`
Header	YAML frontmatter – tools, model, handoffs
Body	Markdown instructions for the AI
Best practice	Be specific, set boundaries, keep concise
Handoffs	Chain agents together for guided workflows

Questions?

Resources:

- VS Code Custom Agents Docs
- GitHub Custom Instructions Docs
- Awesome Copilot Examples